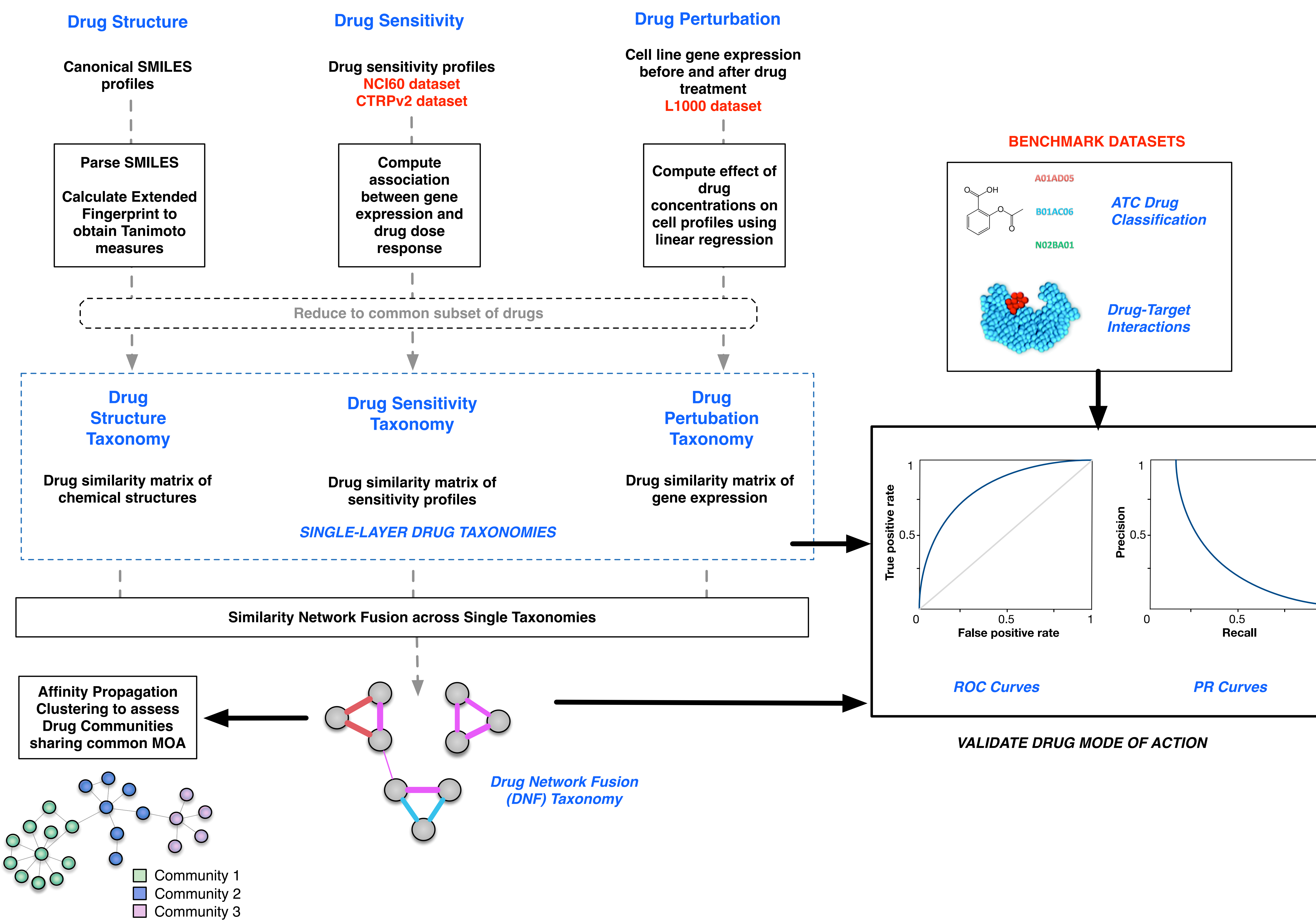
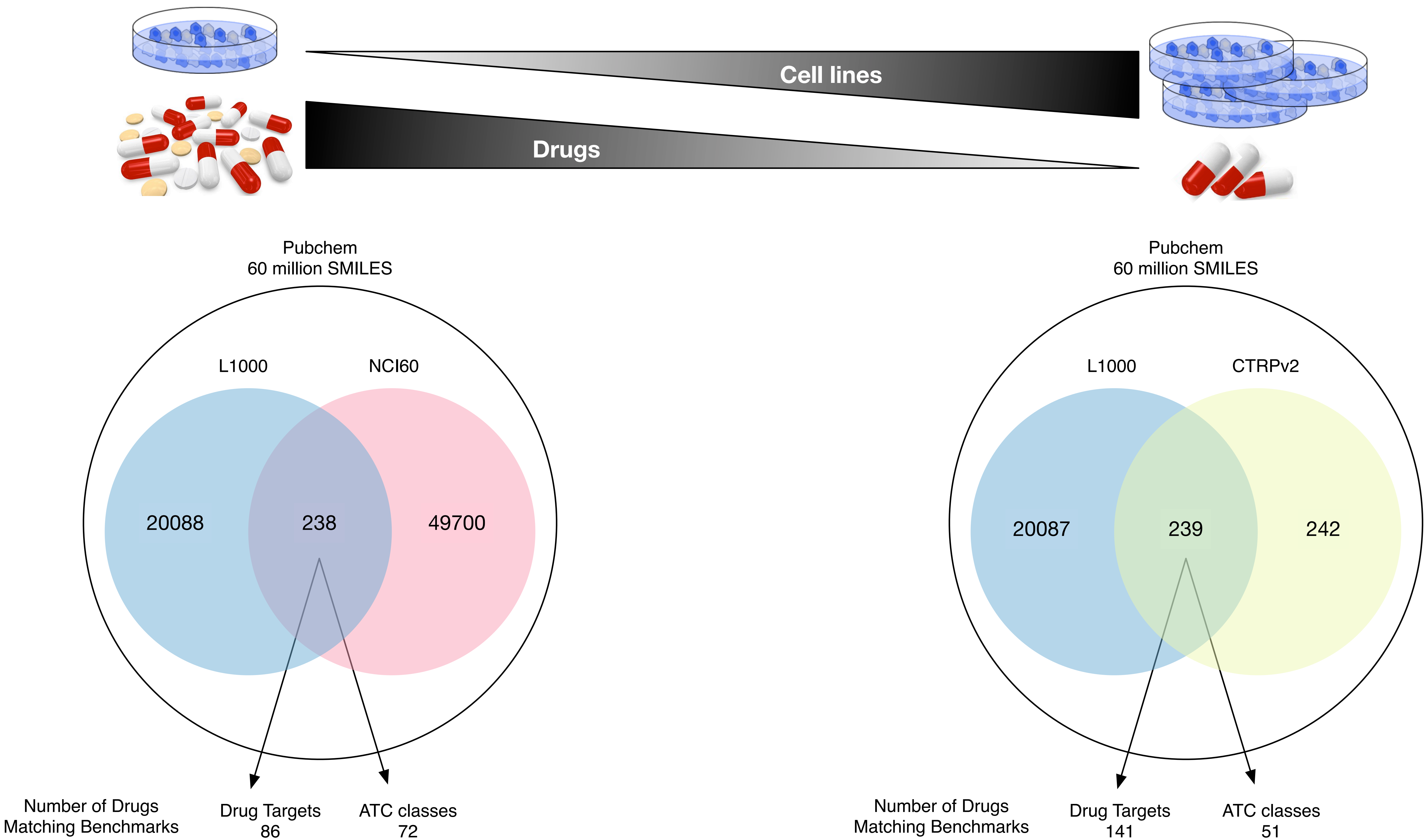
SUPPLEMENTARY FIGURE 1



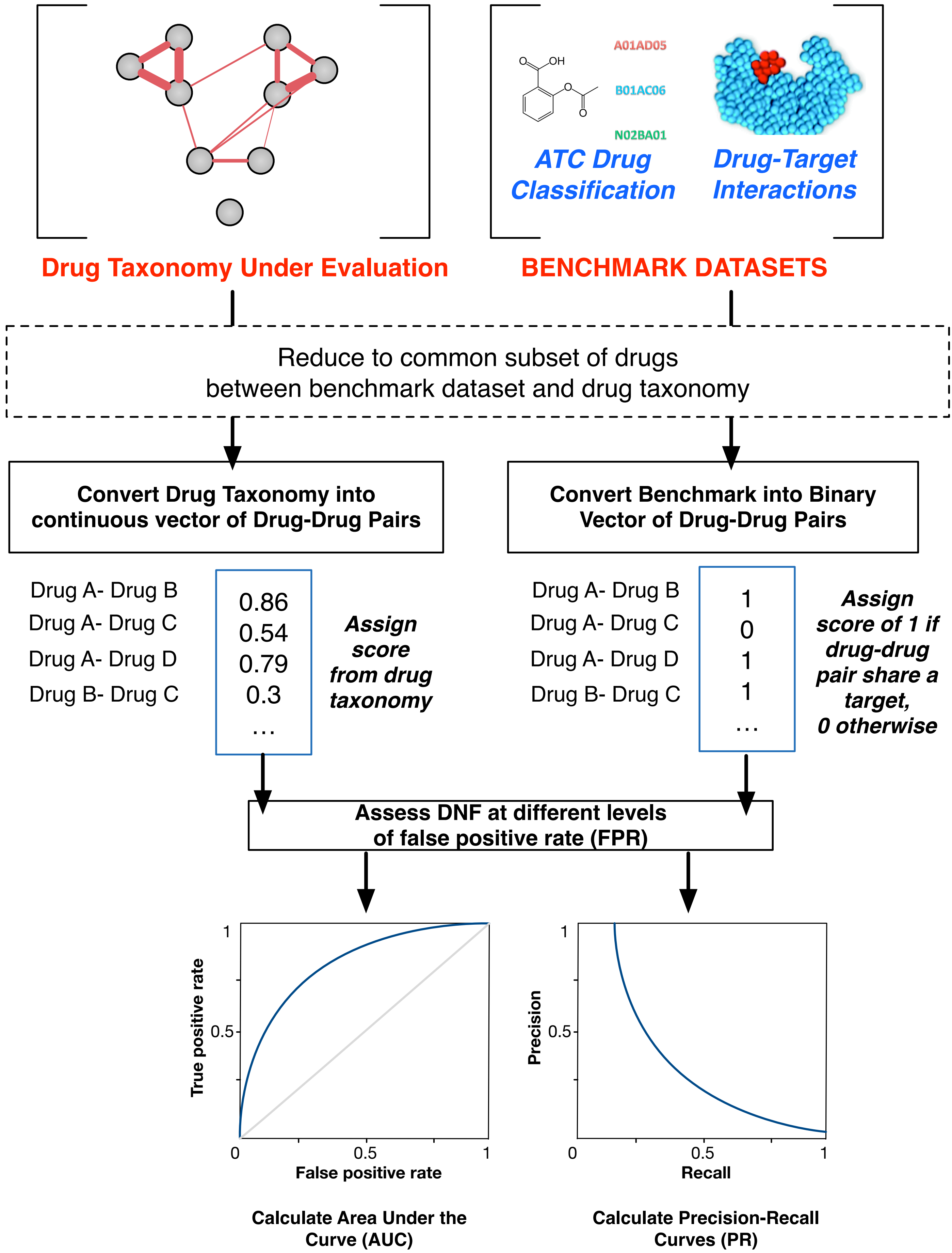
Supplementary Figure 1: Overview of the study design. Drug sensitivity profiles from the NCI60 and the CTRPv2 datasets, along with drug perturbation and drug structure data from the L1000 dataset, are first parsed into drug-drug similarity matrices that represent single-dataset drug taxonomies. Two DNF taxonomies are generated using the drug sensitivity taxonomy from either the NCI60 or CTRPv2 datasets. DNF taxonomies and single-dataset taxonomies are tested against benchmarked datasets containing ATC drug classification and drug-target information, to validate their efficacy in predicting drug MoA. Additional clustering is conducted on DNF taxonomies to identify drug communities sharing a MoA.

SUPPLEMENTARY FIGURE 2



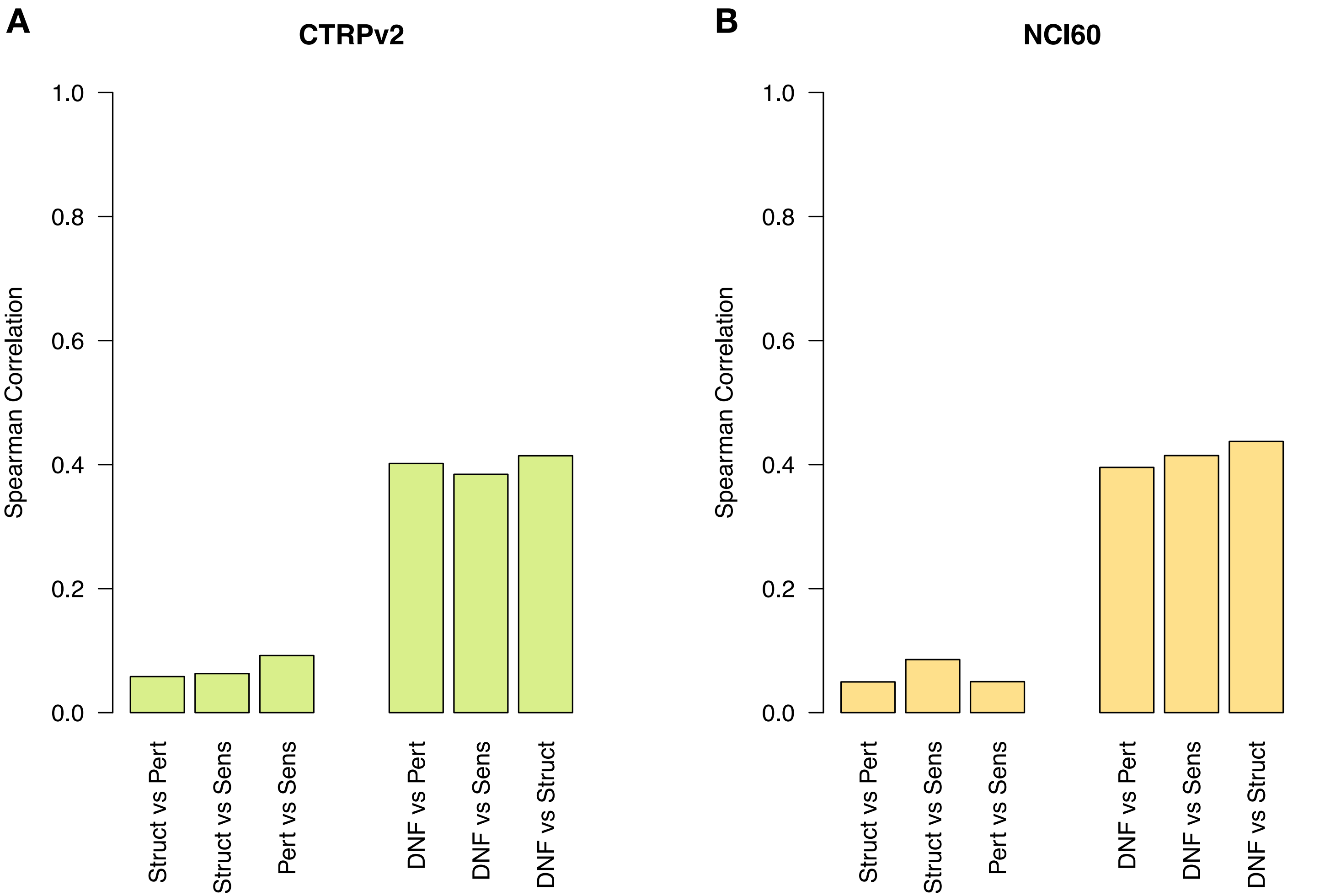
Supplementary Figure 2: Overlap of drug annotations across the L1000 and the NCI60 and CTRPv2 sensitivity datasets. Also indicated are the number of drugs from each DNF matrix, which overlap with the drug target and ATC benchmarks.

SUPPLEMENTARY FIGURE 3



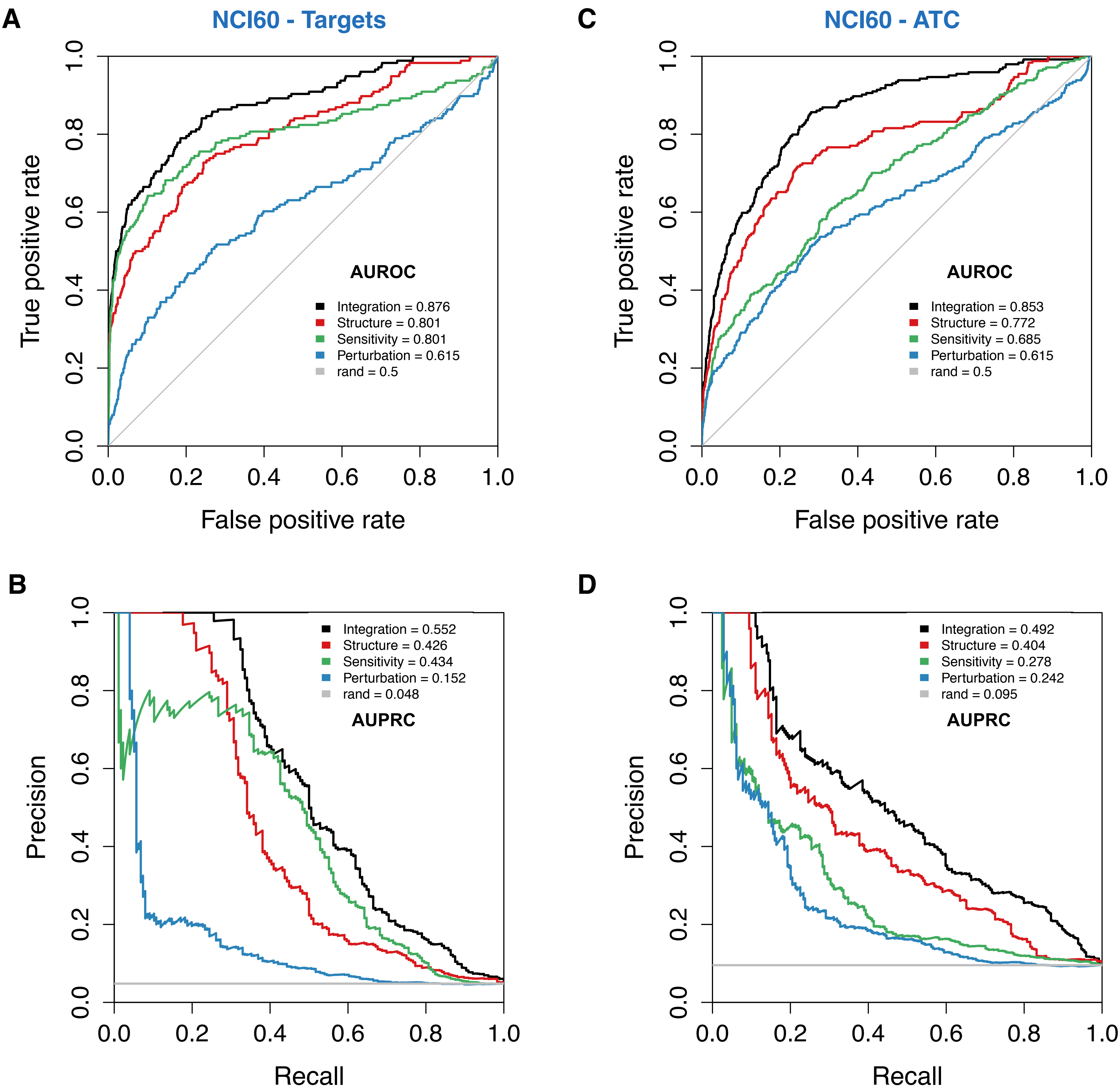
Supplementary Figure 3: Schematic representation of the validation of the DNF and single data type analyses against drug benchmarks. Drug taxonomies are converted into a continuous vector of drug-drug pairs. Benchmark datasets are converted into binary vectors, whereby a given drug-drug pair is assigned a value of ‘1’ if the drugs share a common drug target or ATC classification, and ‘0’ otherwise. Vectors are compared using AUROC and AUPRC.

SUPPLEMENTARY FIGURE 4



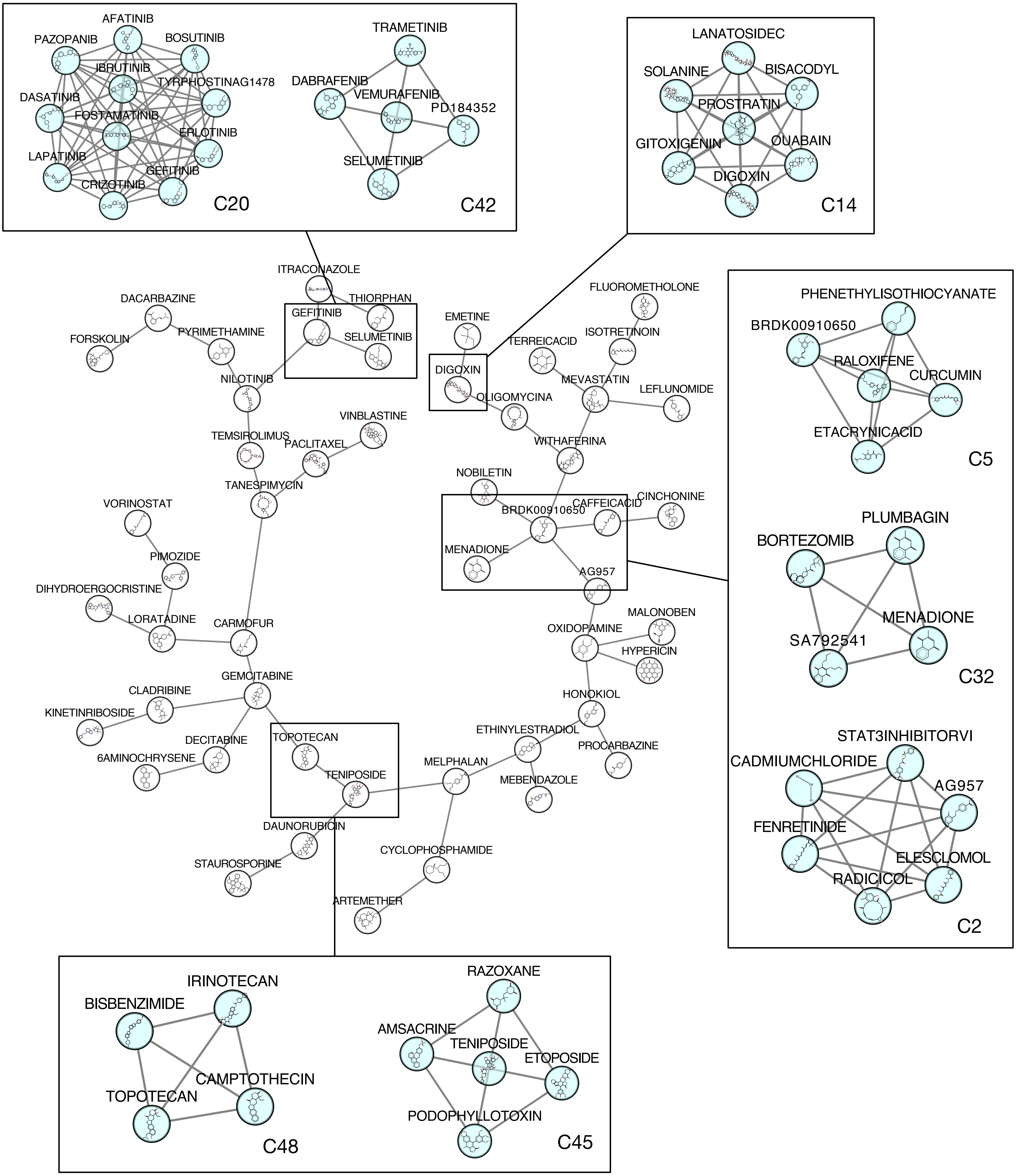
Supplementary Figure 4: Complementarity of drug information across drug taxonomies. Spearman correlation between all pairs of single-layer similarity matrices (drug structure, drug perturbation, drug sensitivity) are depicted. Correlations between the integrative drug taxonomy (DNF) and each of the single-layer similarity matrices are also show. Data are shown for both (**A**) drug taxonomy using CTRPv2 and (**B**) drug taxonomy using the NCI60 sensitivity datasets.

SUPPLEMENTARY FIGURE 5



Supplementary Figure 5: Validation of single-dataset and DNF taxonomies against drug benchmark datasets, based on DNF generated using NCI60. ROC and PR curves are shown for each of the taxonomies, tested against ATC annotations and drug-target information from Chembl or internal benchmarks. A diagonal (grey) representing the null case (AUROC=0.5) is drawn for reference, and a grey curve is also drawn to map random (rand) cases for the PR curves. (**A**) ROC curve for NCI60 against drug-targets (**B**) PR curve against drug-targets (**C**) ROC curve for NCI60 against ATC (**D**) PR curve against ATC drug classifications.

SUPPLEMENTARY FIGURE 6

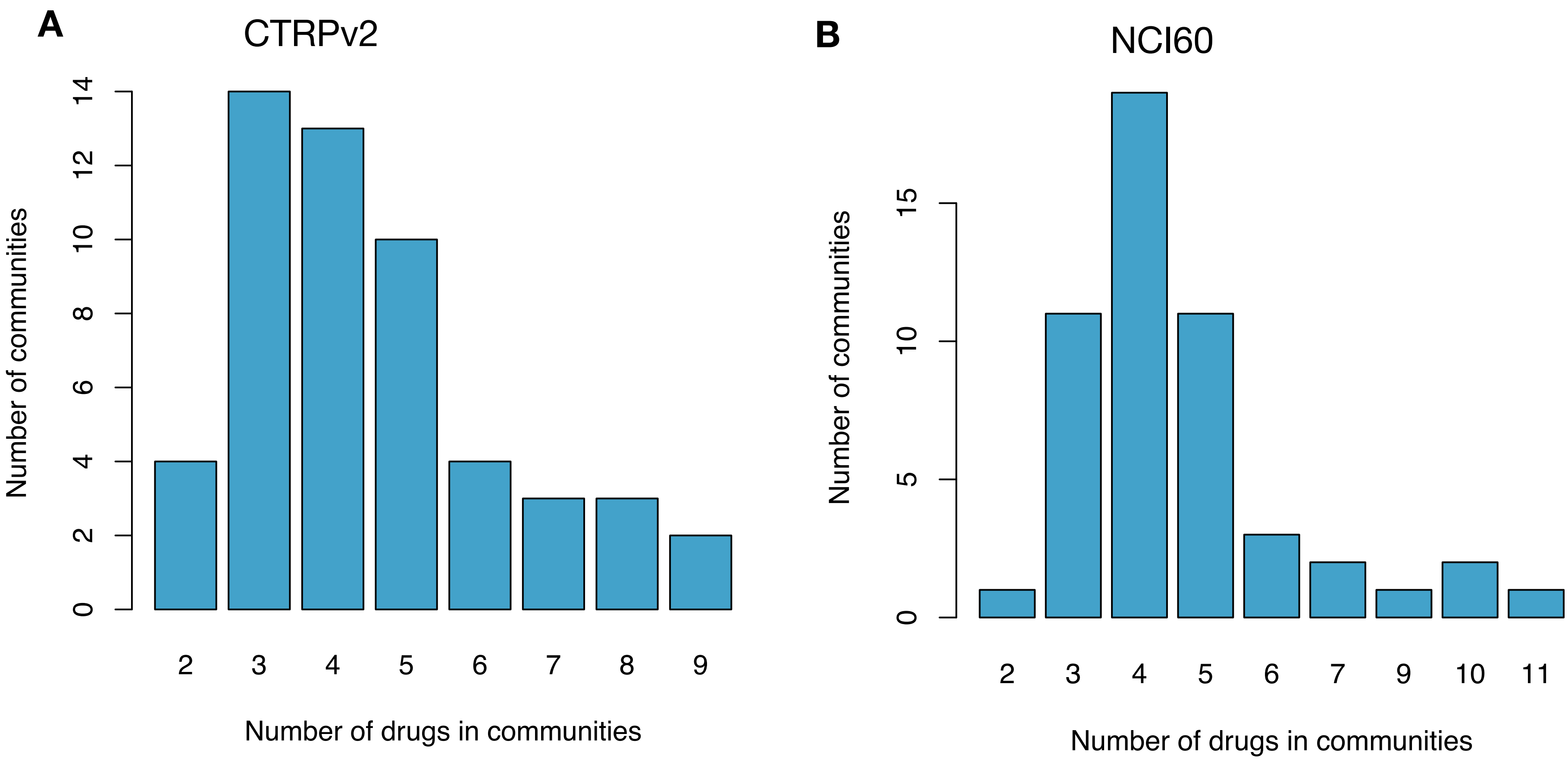


Supplementary Figure 6: Community of 53 Exemplar drugs of the DNF taxonomy generated using NCI60. Communities sharing similar MoA and proximity in the network are highlighted, with the community number indicated.

SUPPLEMENTARY FIGURE 7

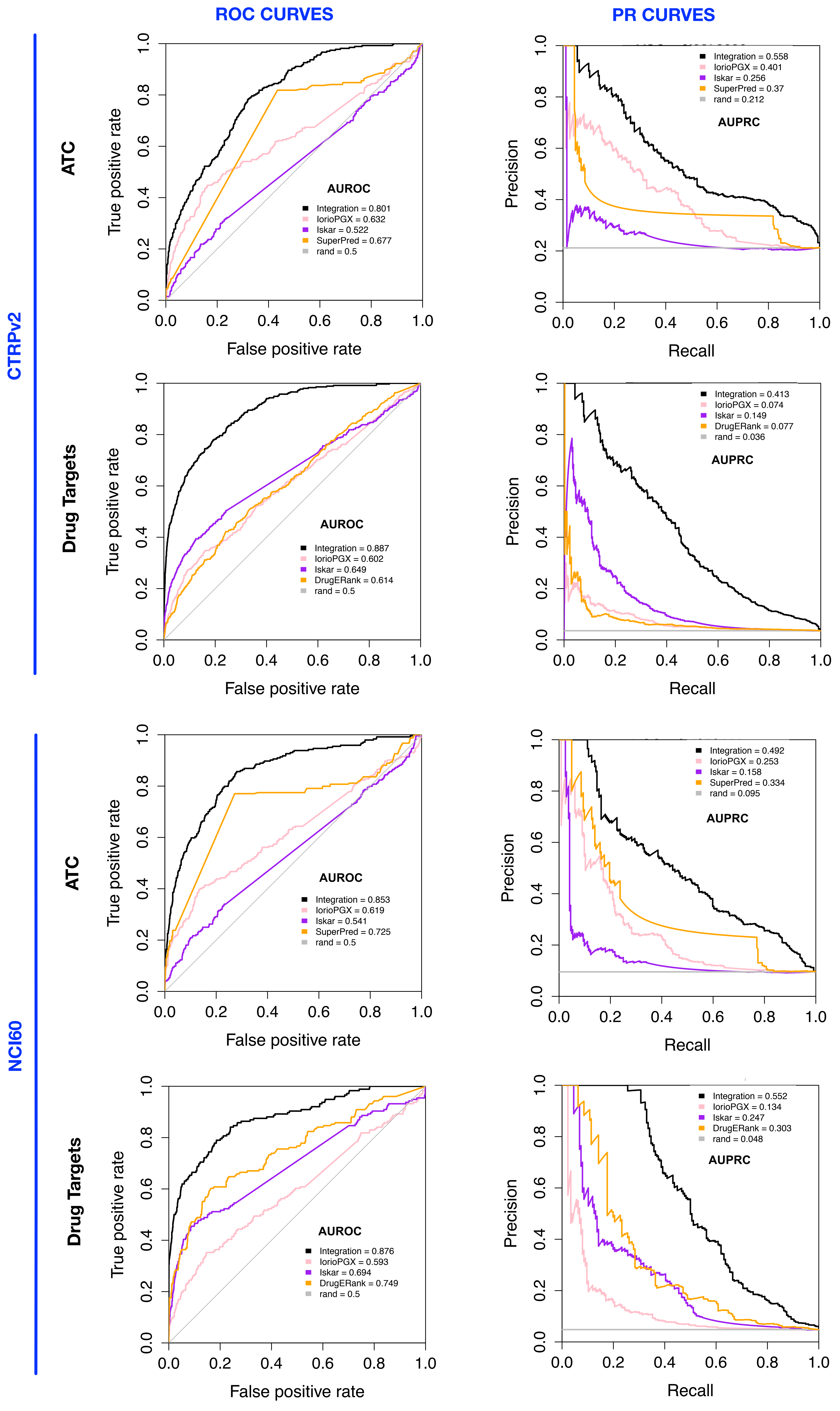
Supplementary Figure 7: Enrichment of Drug Communities of the DNF taxonomy generated using NCI60. A total of 53 communities were tested for enrichment against drug target annotations from DrugBank and ATC annotations from ChEMBL. (**A**) Enrichment of communities for Drug target annotations, with -log10 values indicated in the heatmap, which has been reduced to show significantly enriched communities. Communities are labelled by community number as determined by the APC algorithm. (**B**) Enrichment of communities for ATC classes, with -log10 values indicated in the heat map, which has been reduced to show significantly enriched communities. Communities are labelled by community number as determined by the APC algorithm.

SUPPLEMENTARY FIGURE 8



Supplementary Figure 8: Distribution of drug communities sizes of the DNF taxonomy. (**A**) A total of 53 communities from DNF (using CTRPv2 sensitivity data) were tested for enrichment against drug target annotations from the CTRPv2 data and ATC annotations from ChEMBL (see methods). (**B**) A total of 53 communities from DNF (using NCI60 sensitivity data) were tested for enrichment against drug target annotations from DrugBank and ATC annotations from ChEMBL.

SUPPLEMENTARY FIGURE 9



Supplementary Figure 9: Comparison of the DNF and single-layer drug taxonomies with comparable drug prediction methods (SuperPred, DrugE-Rank, Iorio, Iskar). ROC and PR curves are depicted to indicate the performance of the taxonomies and comparative methods against drug target and ATC benchmarks. This analysis is conducted for both DNF taxonomies (based on CTRPv2 or NCI60 data, shown in blue), and their associated similarity networks. ROC curves are on the left, and PR curves on the right.